Homework 1. Running pipelines

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1. Install nextflow
   use manual: https://www.nextflow.io/docs/latest/install.html

```
(nf-env) polinam@polina:~$ nextflow -v
nextflow version 25.10.2.10555
```

2. Run rnaseq-nf pipeline with engine=docker in conda env

```
(nf-env) polinam@polina:~$ nextflow run nextflow-io/rnaseq-nf -profile docker
Nextflow 25.10.4 is available - Please consider updating your version to it

N E X T F L O W  ~  version 25.10.2

Pulling nextflow-io/rnaseq-nf ...
 downloaded from https://github.com/nextflow-io/rnaseq-nf.git
Launching `https://github.com/nextflow-io/rnaseq-nf` [sleepy_heisenberg] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/polinam/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/polinam/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[49/b4f7a5] RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✔
[16/7d3193] RNASEQ:FASTQC (ggal_gut)                [100%] 1 of 1 ✔
[35/ae1a35] RNASEQ:QUANT (ggal_gut)                 [100%] 1 of 1 ✔
[97/88b257] MULTIQC                                 [100%] 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 06-Apr-2026 23:32:36
Duration    : 1m 42s
CPU hours   : 0.1
Succeeded   : 4
```

```
(nf-env) polinam@polina:~$ nextflow run nextflow-io/rnaseq-nf -profile conda
Nextflow 25.10.4 is available - Please consider updating your version to it

N E X T F L O W  ~  version 25.10.2

Launching `https://github.com/nextflow-io/rnaseq-nf` [sick_gates] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/polinam/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/polinam/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[e3/7ddb78] RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✔
[57/b6b2e8] RNASEQ:FASTQC (ggal_gut)                [100%] 1 of 1 ✔
[19/ef5b95] RNASEQ:QUANT (ggal_gut)                 [100%] 1 of 1 ✔
[1d/247f4e] MULTIQC                                 [100%] 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 06-Apr-2026 23:43:47
Duration    : 1m 41s
CPU hours   : (a few seconds)
Succeeded   : 4
```

3. Test on different RNA-seq data
   Download reads and reference

```
wget ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR633/SRR633726/SRR633726_1.fastq.gz

wget ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR633/SRR633726/SRR633726_2.fastq.gz

wget ftp://ftp.ensemblgenomes.org/pub/plants/release-40/fasta/arabidopsis_thaliana/dna/Arabidopsis_thaliana.TAIR10.dna.toplevel.fa.gz
```

```
(nf-env) polinam@polina:~/hw1-running-pipelines$ nano .nextflow.log
(nf-env) polinam@polina:~/hw1-running-pipelines$ ls
Arabidopsis_thaliana.TAIR10.dna.toplevel.fa  SRR633726_1.fastq.gz  SRR633726_2.fastq.gz  res  work
(nf-env) polinam@polina:~/hw1-running-pipelines$ nextflow run nextflow-io/rnaseq-nf \
  --reads "$PWD/SRR633726_{1,2}.fastq.gz" \
  --transcriptome "$PWD/Arabidopsis_thaliana.TAIR10.dna.toplevel.fa" \
  --outdir results \
  -profile docker
Nextflow 25.10.4 is available - Please consider updating your version to it

N E X T F L O W  ~  version 25.10.2

Launching `https://github.com/nextflow-io/rnaseq-nf` [boring_euclid] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/polinam/hw1-running-pipelines/Arabidopsis_thaliana.TAIR10.dna.toplevel.fa
reads        : /home/polinam/hw1-running-pipelines/SRR633726_{1,2}.fastq.gz
outdir       : results

executor >  local (4)
[5b/6de527] RNASEQ:INDEX (Arabidopsis_thaliana) [100%] 1 of 1 ✔
[86/e0d1ed] RNASEQ:FASTQC (SRR633726)           [100%] 1 of 1 ✔
[7f/05d394] RNASEQ:QUANT (SRR633726)            [100%] 1 of 1 ✔
[ba/d190f7] MULTIQC                             [100%] 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 07-Apr-2026 00:05:59
Duration    : 3m 36s
CPU hours   : 0.1
Succeeded   : 4
```